Docker CMD vs EntryPoint

Run docker command from ubuntu image, it runs the instance of ubuntu image, and exits immediately.

```
docker run ubuntu
```

To get list of running containers:

```
docker ps
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS
```

List containers including that are stoped/exited:

```
docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS
45aacca36850        ubuntu              "/bin/bash"         43 seconds ago      Exited (0) 41 seconds ago
```

Unlike virtual machines, containers are not meant to host operating systems, containers are meant to run specific task, process, to host instance of web server, application server, database, carry out computation, and analysis.
Once the task completes the container exits.
The container only lives, until the process inside it is alive.
If the webservice inside is stopped or crashes, the container exits.
Who defines what process to run within the container?
If you look at popular docker images like nginx,

Which as a instruction,

```
# Define default command.
CMD ["nginx"]
```

like CMD [“nginx”], that defines the program to run, within the container, when it starts. For the nginx images, its the nginx command.

For mysql image its “mysqld” command.

```
CMD ["mysqld"]
```

```
# Install Nginx.
RUN \
  add-apt-repository -y ppa:nginx/stable && \
  apt-get update && \
  apt-get install -y nginx && \
  rm -rf /var/lib/apt/lists/* && \
  echo "\ndaemon off;" >> /etc/nginx/nginx.conf && \
  chown -R www-data:www-data /var/lib/nginx

# Define mountable directories.
VOLUME ["/etc/nginx/sites-enabled", "/etc/nginx/certs", "/etc/nginx/con

# Define working directory.
WORKDIR /etc/nginx

# Define default command.
CMD ["nginx"]
```

```
ARG MYSQL_SERVER_PACKAGE_URL=https://repo.mysql.com/yum/mysql-8.0-community/docker/x86_
ARG MYSQL_SHELL_PACKAGE_URL=https://repo.mysql.com/yum/mysql-tools-community/el/7/x86_

# Install server
RUN rpmkeys --import https://repo.mysql.com/RPM-GPG-KEY-mysql \
  && yum install -y $MYSQL_SERVER_PACKAGE_URL $MYSQL_SHELL_PACKAGE_URL libpwquality \
  && yum clean all \
  && mkdir /docker-entrypoint-initdb.d

VOLUME /var/lib/mysql

COPY docker-entrypoint.sh /entrypoint.sh
COPY healthcheck.sh /healthcheck.sh
ENTRYPOINT ["/entrypoint.sh"]
HEALTHCHECK CMD /healthcheck.sh
EXPOSE 3306 33060
CMD ["mysqld"]
```

For the Ubuntu:

```
# Pull base image.
FROM ubuntu:14.04

# Install.
RUN \
  sed -i 's/# \(.*multiverse$\)/\1/g' /etc/apt/sources.list && \
  apt-get update && \
  apt-get -y upgrade && \
  apt-get install -y build-essential && \
  apt-get install -y software-properties-common && \
  apt-get install -y byobu curl git htop man unzip vim wget && \
  rm -rf /var/lib/apt/lists/*

# Add files.
ADD root/.bashrc /root/.bashrc
ADD root/.gitconfig /root/.gitconfig
ADD root/.scripts /root/.scripts

# Set environment variables.
ENV HOME /root

# Define working directory.
WORKDIR /root

# Define default command.
CMD ["bash"]
```

Where bash is the default command.
Bash is not like a process, like web server or database server, it is a shell that listens input from the terminal, if it not finds the terminal it exits.

When we run the ubuntu container earlier, docker created a container from ubuntu image, and lost the bash program, by default docker does not attached terminal to container when it is run, so the bash program does not find the terminal and so it exits.

Since the process that was started, when the container is created is finished, the container exits as well.

So how do you specify the different command to start the container.
Append a command to docker run command, so it overrides the default command that is specified for the image. Example:

```
docker run ubuntu [COMMAND]

docker run ubuntu sleep 5
```

sleep for 5 seconds and terminates.

How to make that change permanent, want the image to start the sleep command when it starts,

```
FROM Ubuntu

CMD sleep 5
```

```
CMD command param1
CMD ["command", "param1"]

CMD sleep 5
CMD ["sleep", "5"]
CMD ["sleep 5"]
```

```
docker build -t ubuntu-sleeper .

docker run ubuntu-sleeper
```

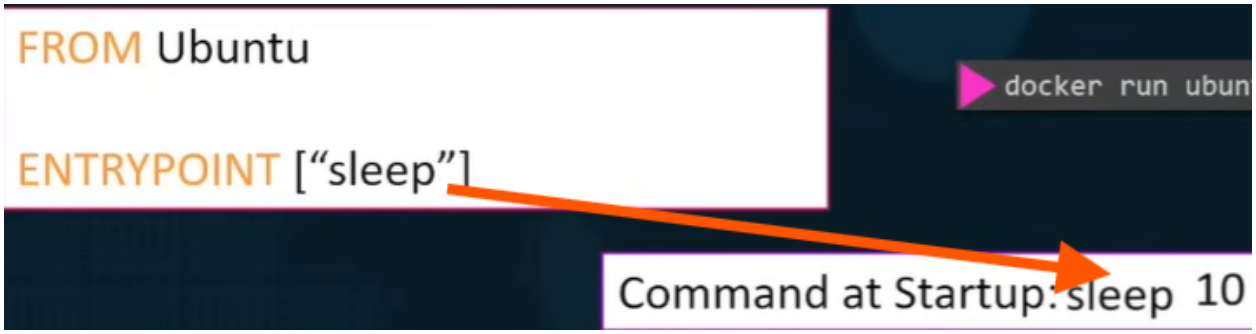
Want to wish to change the sleep, by appending to docker run command:

```
docker run ubuntu-sleeper sleep 10
```

We wish to provide only number of seconds to sleep, and dont want to provide sleep command again. Like

```
docker run ubuntu-sleeper 10
```

that ‘s where entrypoint comes into play,

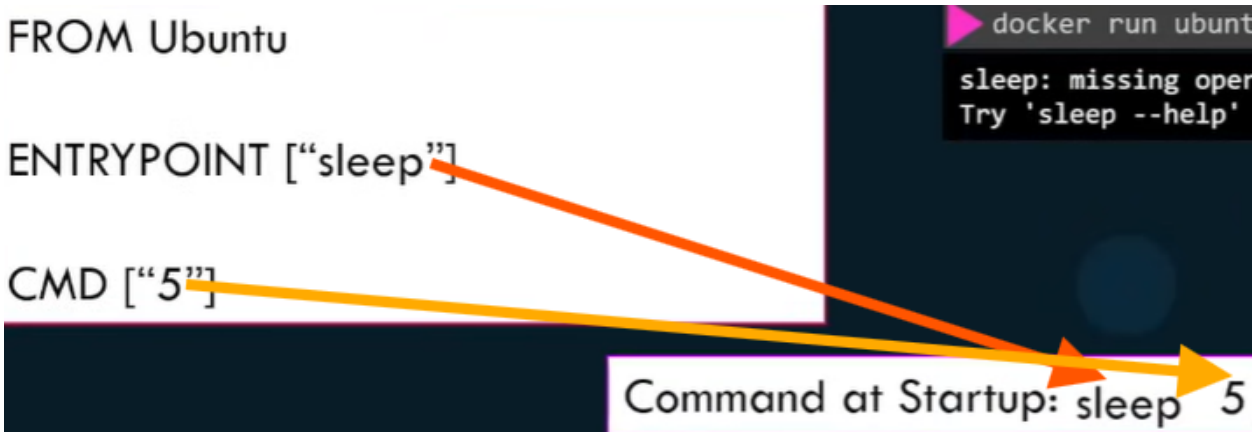



```
docker run ubuntu-sleeper 10
```

What if we dont provide the number:

```
docker run ubuntu-sleeper
sleep: missing operand
Try 'sleep --help' for more information.
```

So there has to be default value if not provided: that's where command instruction is specified.



To modify the entrypoint during runtime: from sleep to sleep2.0